HYDRA Documentation

HYDRA MW4.0pe

Preparation Guide for SLES12 SP5

Version 1.0.23049

Last changed on: 02.09.2020

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Contents

1 Introduction

This configuration manual explains how to configure SUSE Linux Enterprise Server 12 SP5 (SLES12 SP5) for use as a server for MPDVs Manufacturing Integration Platform (**MIP**) based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1.

The configuration of the server should always be based on MPDVs hardware and software recommendations for the respective MIP based product.
See manuals: **HW_SW_GUIDE.pdf**

!ATTENTION!
The Server must be a dedicated server for running any MIP based products.
There must be no additional functionality used, e.g. like File Server, Print Server, OPC Server, Domain Controller (PDC, BDC), Active Directory Controller or similar.

If the configuration of your server deviates from the configuration described in this manual, additional efforts and expenses might be necessary for the installation of MIP based products.
Already agreed shipping or installation appointments might be delayed or may have to be postponed.

MIP based products might not be able to work properly on such a server.
It might even be impossible to perform a successful installation on such a server.

It might not be possible to install different MIP based products on the same server.

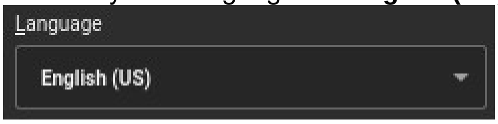
2 Operating System

- Server platform according to the hardware and software recommendations of MPDV Mikrolab GmbH is required.

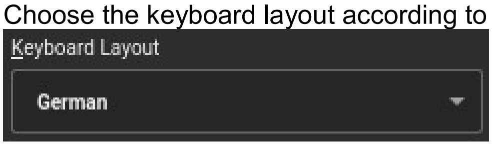
- The version of your operating system must meet the recent recommendations of MPDV Mikrolab GmbH. For the time being that is:
  SUSE Linux Enterprise Server 12 SP5 (x86_64) (SLES12 SP5)

- The server must be a **dedicated server** for running any MIP based product.
  There must be no additional functionality be used, e.g. like File Server, Mail Server, Print Server, OPC Server, Domain Controller (PDC, BDC), Active Directory Controller or similar.
  In such cases it might be impossible to perform a successful installation of MIP based products.

- Set the system language as "**English (US)**":

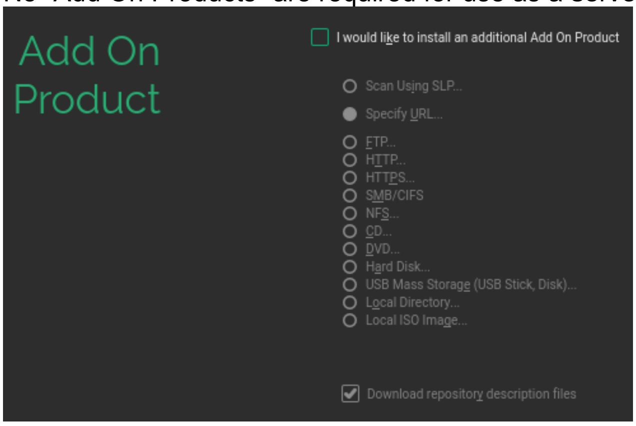


- Choose the keyboard layout according to your local environment, e.g. "German":



- Agree to the License Terms:

- No "Add On Products" are required for use as a server for MIP based products:



- **System Role**

  To install the Oracle database software a graphic X-Window user interface (e.g. **KDE** or **GNOME**) is mandatory. GNOME is the default desktop environment of SLES12.

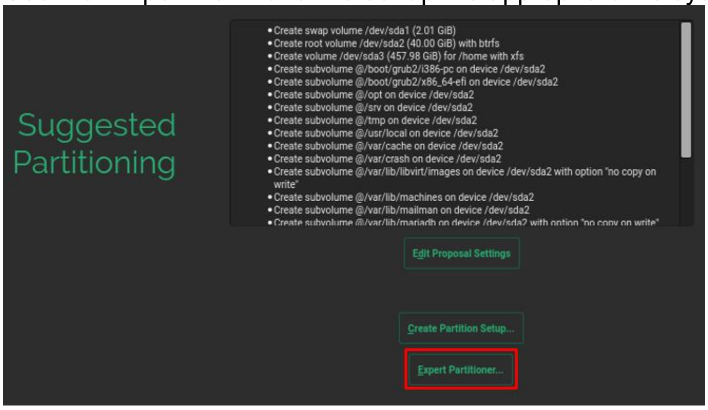
  Install system role "Default System":

- **Suggested Partitioning**

  For information about disk requirements and partitioning see chapter "3 Hard Disk Layout".

  All partitions except the swap partition must be formatted with file system type "**ext3**".

  Use the "Expert Partitioner" to set up the appropriate file systems:



- **File Systems**

  All partitions except the swap partition must be formatted with file system type "**ext3**".

  e.g.:

Device	Size	F	Enc	Type	FS Type	Label	Mount Point
/dev/sda1	100.00 GiB	F		Linux native	Ext3		/
/dev/sdb1	600.00 GiB	F		Linux native	Ext3		/u1

- **Swap Partition**

  Use file system type "**swap**"

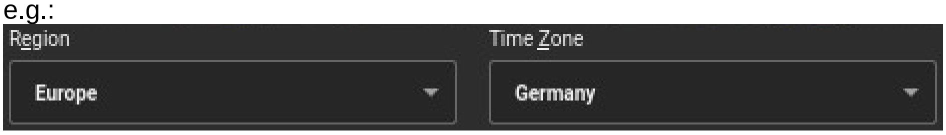
  The **size** of the swap partition must be at least **2 times** of the size of the available amount of **RAM**, e.g.: 64 GB swap space for 32 GB RAM

Device	Size	F	Enc	Type	FS Type	Label	Mount Point
/dev/sda2	64.00 GiB	F		Linux swap	Swap		swap

- **Region and Time Zone**

  Set Region and Time Zone according to the location of your server,

  e.g.:



- **Date and Time**

  Set checkbox for "**Hardware Clock Set to UTC**"

  Set Date and Time according to your local date and time.

- **Local User "mipadm"**

  The password for user "mipadm" should be set to "**Mip74821**".

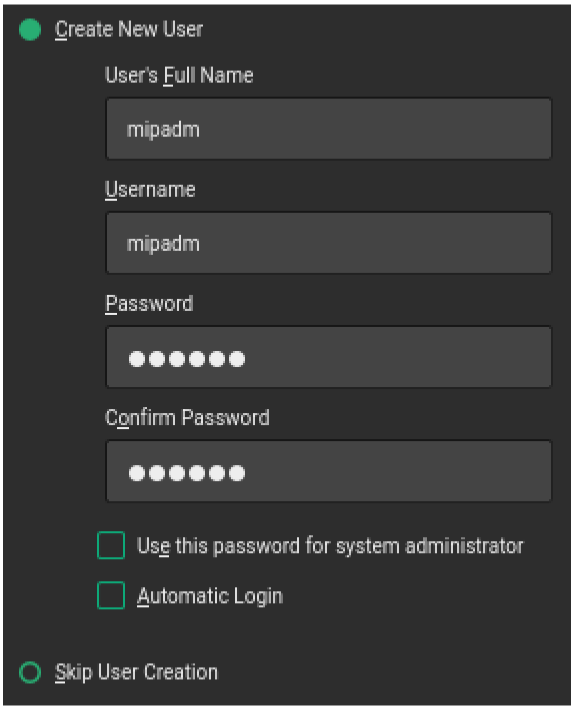
  Optionally any other password can be used which must then be disclosed to the MPDV employees who are installing the MIP based product for you.

  The following characters are **not** allowed for the password:

  - Characters with ASCII Code **> 126** (e.g. German umlauts, French "umlauts", etc.)

  - Pipe**: |**

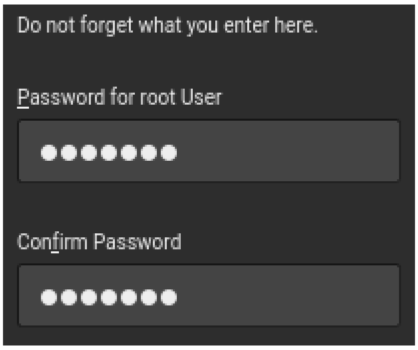
  - Ampersand: **&**



- **User “root”**

  The password for user "**root**" is free to choose.

  Be prepared to disclose that password to the MPDV employees who are installing the MIP based product for you.



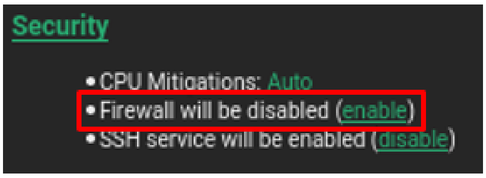
- **Software packages for SLES12**

  Additional to the default software packages of SLES12 add the following packages:

  "**C/C++ Compiler and Tools**" and "**Oracle Server Base**":

- **Firewall**

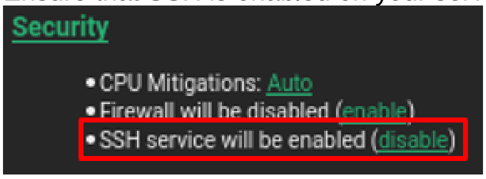
  The local firewall must be disabled.



- **OpenSSH** is the required SSH software for Oracle Databases.

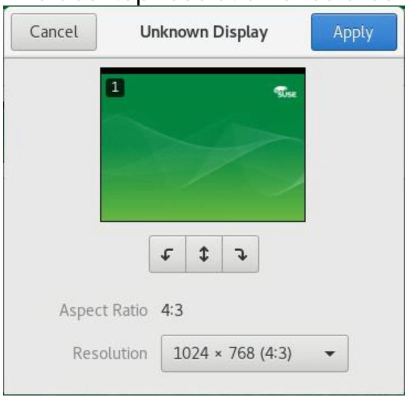
  The SSH implementation coming with SUSE Linux Enterprise Server is OpenSSH.

  Ensure that SSH is enabled on your server.



- **Desktop Resolution**

  The desktop resolution should be set at least to **1024x768**.



  The monitor must be configured to display at least **256 colors**.

- **Network Configuration**

  TCP/IP (IPv4) network protocol based on a Ethernet network must be available.

  At least the following TCP/IP ports and port ranges must be available for MIP based products:

  1521, 3300-3399, 3299, 8080, 9000-9005, 10000, 10100, 10103, 10111, 10120-10127, 10150, 10177, 18080, 30101-30108

  If these ports (especially port 10000 for HYDRA) are not available there might be additional efforts necessary during the installation.

  Additional ports will be required when the MIP based product is supposed to be installed as a multi system installation (e.g. HYDRA) or when additional functionality should be used.

- **IP address** and **hostname** are free to choose.

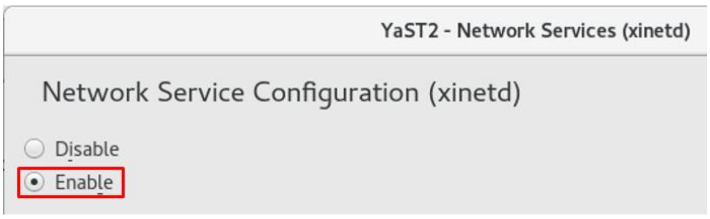
  It is mandatory that the MIP based product server uses a **dedicated IP address**.

  It is not possible for the MIP server to receive its IP address from a DHCP server.

  All other settings like subnet mask, name server, router or gateway, etc. must be set according to your local network configuration.

- **xinetd**

  The "Network Service Configuration (**xinetd**)" must be enabled:



- **telnet, ftp**

  Depending on the system environment it might be necessary that one or more of the following services must be activated in the **Network Service Configuration (xinetd)**:

  telnet (not necessary if connection type SSH (=Default) is to be used)

Status	Service	Type	Protocol	Wait	User	Server
On	telnet	stream	tcp	No	root	/usr/sbin/in.telnetd

  To activate telnet the installation-DVD, e.g. "SLE-12-SP5-Server-DVD-x86_64-GM-DVD1.iso", must be available at the server to install the additional software package “telnet-server”.

  ftp (required for exchanging files between MIP based products and 3rd party systems like ERP, DNC, etc.)

Status	Service	Type	Protocol	Wait	User	Server
On	ftp	stream	tcp	No	root	/usr/sbin/vsftpd

- Modify file **/etc/vsftpd.conf**

  Check that the following settings are enabled:

  ```
  write_enable=YES
  local_enable=YES
  local_umask=022
  listen=NO
  ```

  Disable “listen_ipv6”:

  ```
  # listen_ipv6=YES
  ```

- Modify file **/etc/ftpusers**

  Enable FTP access for user oracle:

  ```
  # oracle
  ```

- **Group mip**

  A new group “**mip**” must be created.

  The group ID (gid) can be set at your discretion.

- **Group dba**

  In SLES12 the group “**dba**” should be available as default (set filter: “System Groups”).

  If not, create it. The group ID (gid) can be set at your discretion.

- **User mipadm**

  Either create a new local user account “**mipadm**” (if not done already during the software installation, see above) or edit the settings of the already existing user account “mipadm”.

  The user ID (uid) can be set at your discretion.

  The password for user “**mipadm**” is free to choose.

  Be prepared to disclose that password to the MPDV employees who are installing the MIP based product for you.

  Set “**Home Directory**” to **/u1/mip1/sys** (assuming the hard disk layout matches the recommendations in chapter “3 Hard Disk Layout”).

  The “**Login Shell**” must be set to **/bin/mksh**.

  The user’s “**Default Group**” must be changed to "**mip**".

- **.profile**

  Add the following lines to the shell configuration file (/u1/mip1/sys/.profile) of user "mipadm":

  ```
  HN=`uname -n`
  export PS1='$HN:$LOGNAME:$PWD> '
  set -o vi
  ```

- **User oracle**

  In SLES12 there is a disabled user account “**oracle**” available as default (set filter: “System Users”).

  The user account “**oracle**” must be enabled.

  If there might be no such user account, create it. The user ID (uid) can be set at your discretion.

  The password for user “**oracle**” is free to choose.

  Be prepared to disclose that password to the MPDV employees who are installing the MIP based product for you.

  Set “**Home Directory**” to **/home/oracle** (must not coincide with $ORACLE_BASE, e.g. /u1/oracle)

  The “**Login Shell**” must be set to **/bin/mksh**.

  The user’s “**Default Group**” must be changed to "**dba**".

  Under “**Additional Groups**” add “**oinstall**” and “**dba**” as well.

- **.profile**

  Create the shell configuration file /home/oracle/.profile for the user “oracle” and add the following lines:

  ```
  umask 022
  HN=`uname -n`
  export PS1='$HN:$LOGNAME:$ORACLE_SID:$PWD> '
  set -o vi
  cd $ORACLE_BASE
  ```

- **System Console**

  All users must have write permission on the system console:

  `chmod 666 /dev/console`

- **Tape Drive** (if available)

  Access to the tape device (e.g.: **/dev/st0** or **/dev/rmt0**) must be available for all user accounts, e.g.:

  `chmod 777 /dev/st0`

- **Message Queues**

  The amount of available message queues must be **> 64**.

  Check with command "**`ipcs -l`**":

  ```
  ------ Messages Limits --------
  max queues system wide = 32000
  ```

- **System Activity**

  For use of the system activity report tools **sar** and **iostat** the software package **sysstat** is necessary (in SLES12 available by default).

- **Kernel Parameter**

  The necessary kernel parameters are set by the package "**orarun**" provided by "**Oracle Server Base**". Therefore the following files need to be configured:

  /etc/sysconfig/oracle

  ```
  ORACLE_BASE=/u1/oracle
  START_ORACLE_DB="yes"
  START_ORACLE_DB_LISTENER="yes"
  START_ORACLE_DB_EMANAGER="yes"
  SHMMAX= Recommended: SHMMAX = 0.5*(physical memory)
  ```

  example for 24GB RAM: `SHMMAX=12884901888`

  do not use values smaller than `SHMMAX=4294967295`

  /etc/profile.d/oracle.sh

  ```
  ORACLE_BASE=/u1/oracle
  ORACLE_HOME=$ORACLE_BASE/ora19
  ORACLE_SID=MIP1
  ```

  Check that the following settings are disabled:

  ```
  # export ORA_CRS_HOME=$ORACLE_BASE/product/12cR1/crs
  # export ORA_ASM_HOME=$ORACLE_BASE/product/12cR1/asm
  ```

  /etc/profile.d/oracle.csh

  ```
  setenv ORACLE_BASE /u1/oracle
  setenv ORACLE_HOME ${ORACLE_BASE}/ora19
  setenv ORACLE_SID MIP1
  ```

  Check that the following settings are disabled:

  ```
  # setenv ORA_CRS_HOME ${ORACLE_BASE}/product/12cR1/crs
  # setenv ORA_ASM_HOME ${ORACLE_BASE}/product/12cR1/asm
  ```

- Activate the new kernel parameters by running the following command as user “**root**”:

  ```
  cd /etc/init.d
  ./oracle start
  ```

- Check the current size of your shared memory file system /dev/shm (use command **df -h**).

```
Filesystem        Size   Used Avail Use% Mounted on
tmpfs              12G    80K   12G    1% /dev/shm
```

  The shared memory file system /dev/shm must be big enough to accommodate the MEMORY_TARGET and MEMORY_MAX_TARGET values of all installed database instances.
  Otherwise Oracle will throw the following error message:
  "**ORA-00845: MEMORY_TARGET not supported on this system**".

  To adjust the shared memory file system size add the following line to the file "**/etc/fstab**":
  Make sure that the value for "size=" suits your system settings.

```
tmpfs          /dev/shm    tmpfs        size=24g    0 0
```

  Reboot your server afterwards.

- **Software Packages**
  The following software packages need to be installed manually as user root because they are not part of the default installation of SLES12 SP5:

```
yast --install libjpeg-turbo
yast --install libjpeg62
yast --install libjpeg62-turbo
yast --install pixz
yast --install rdma-core
```

  The installation-DVD, e.g. "SLE-12-SP5-Server-DVD-x86_64-GM-DVD1.iso", must be available at the server.

- The latest released versions of the following packages are required by Oracle 19c (see Oracle database installation guide "19c for Linux").
  All packages are part of a default SLES12 SP5 installation except those specially marked.
  To check which packages are installed use "**rpm –qa <package-name>**":

```
bc
binutils
glibc
glibc-devel
libX11
libXau6
libXtst6
libcap-ng-utils
libcap-ng0
libcap-progs
```

libcap1	
libcap2	
libelf-devel	
libgcc_s1	
libjpeg-turbo	(not installed as default in SLES12 SP5, see above)
libjpeg62	(not installed as default in SLES12 SP5, see above)
libjpeg62-turbo	(not installed as default in SLES12 SP5, see above)
libpcap1	
libpcre1	
libpcre16-0	
libpng16-16	
libstdc++6	
libtiff5	
libaio-devel	
libaio1	
libXrender1	
make	
mksh	
net-tools (for Oracle RAC and Oracle Clusterware)	
nfs-kernel-server (for Oracle ACFS)	
pixz	(not installed as default in SLES12 SP5, see above)
rdma-core	(not installed as default in SLES12 SP5, see above)
smartmontools	
sysstat	
xorg-x11-libs	
xz	

2.1 Administration Console PC

With a MIP based product running on a Linux server you must have a dedicated Windows PC available where certain administration tools and certain client software like the MES Operation Center (MOC) or the APS Operation Center (AOC) will be installed.
That Windows PC will serve as access point for remote support sessions by MPDV Mikrolab GmbH.
The Administration Console PC is mandatory for all support and maintenance purposes by MPDV!

The Windows operating system of this PC should meet the requirements for the HYDRA client software MES Operation Center (MOC), e.g. Windows 10 64bit.
For more detailed information please see the hardware and software recommendations of MPDV Mikrolab GmbH: **HW_SW_GUIDE.pdf**

2.2 Web Browser Software

If you plan to use MPDV's product Smart MES Applications (SMA) version **8.2** you must be aware that Microsoft's Internet Explorer (IE) is no longer supported by MPDV.
You need to provide an alternative web browser software like **Google Chrome** which is MPDV's recommended web browser software for using SMA 8.2.
Please make sure that Google Chrome is installed on your Administration Console PC.

3 Hard Disk Layout

The storage demand of a MIP based system and its database usually increases over time.
Therefore the available disk storage should be large enough right from the start.

For a MIP based system we recommend the following hard disk capacity to be available:
200 GB for the MIP based application including possible archive data to be generated in the future
400 GB for the MIP based product database (default configuration)

Apart from sufficiently large root (**≥ 50 GB**) and swap (**size = 2 x RAM**) partitions there should be at least another partition with file system **/u1** available which must provide the above mentioned disk capacity of at least **600 GB**.

If there are additional hard disks and partitions available, e.g. like **/u2, /u3, /u4, /u5** etc., the data files for the MIP based product database could be spread for performance reasons.
When using online backup solutions for the database software the use of a dedicated hard disk and file system is highly recommended.

All partitions except the swap partition must be formatted with file system type **ext3**.

Configuration example for a SLES12 server with 5 hard disks (each 500GB in size):

1. hard disk:

/	50 GB	Standard root file system
swap	2 x RAM	Swap Partition
/u1	~350 GB	MIP based product and database application

2. hard disk:

/u2	500 GB	database data files

3. hard disk:

/u3	500 GB	database data files

4. hard disk:

/u4	500 GB	database data files

5. hard disk (additional disk for online backup solutions):

/u5	500 GB	backup files for online backup solutions

4 Directories

Directory **/u1/mip1** must be set to owner **mipadm** and group **mip**

```
chown mipadm:mip /u1/mip1
```

The access permissions must be set to 775, e.g.:

```
chmod 775 /u1/mip1
drwxrwxr-x 3 mipadm mip   4096 Jan 28 16:05 mip1
```

Create a new directory **/u1/export**

```
mkdir /u1/export
```

Directory **/u1/export** must be set to owner **mipadm** and group **mip**

```
chown mipadm:mip /u1/export
```

The access permissions must be set to 775, e.g.:

```
chmod 775 /u1/export
drwxrwxr-x 2 mipadm mip   4096 Jan 28 16:30 export
```

Create new directories **/u1/oracle**, **/u2/oracle**, **/u3/oracle**, **/u4/oracle, /u5/oracle**, etc.

```
mkdir /u1/oracle, etc.
```

All directories **/u1/oracle**, **/u2/oracle**, **/u3/oracle**, **/u4/oracle, /u5/oracle**, etc. must be set to owner **oracle** and group **dba** e.g.:

```
chown oracle:dba /u1/oracle, etc.
```

The access permissions should be set to 755 by default, e.g.:

```
chmod 755 /u1/oracle, etc.
drwxr-xr-x 2 oracle dba    4096 Dec  8 12:39 /u1/oracle
```

Access permission for all directories **/u1, /u2, /u3**, etc. must be set to 777, e.g.:

```
chmod 777 /u1
drwxrwxrwx  14 root root  4096 Dec  1 16:19 u1
```

5 Database ORACLE

For information about currently supported database versions please consult the recent recommendations regarding hardware and software by MPDV Mikrolab GmbH.

If the Oracle database software is supposed to be preinstalled by the customer himself, please contact MPDV Mikrolab GmbH first.
You will then be provided with the proper database installation and configuration manual(s) which will allow you to do the installation and configuration according to the needs of the MIP based product.

If you plan to have separate application and database servers the 64Bit Oracle client software needs to be installed on the MIP application server.

All MIP based product databases are supposed to use the character set **AL32UTF8**.
The use of other character sets is not supported and may cause problems with the MIP based application.

The Archive Log Mode for Oracle databases installed by MPDV Mikrolab GmbH is **deactivated** by default. This will prevent an unnoticed overflow of file systems and therefore the halt of all database functionality. The Archive Log Mode will be activated by MPDV Mikrolab GmbH on special request by the customer only.

When using online backup solutions with an activated Archive Log Mode the use of a dedicated hard disk to store Oracle's archive log files is highly recommended.
See chapter "3 Hard Disk Layout".